General Mathematics — Exercise Sheet 14

Lines, planes and coordinates

Variant: Statements

Exercise 1 (A parameter in the line direction). Let Π have equation $2x + y - z = 3$. For each $a \in \mathbb{R}$, classify the intersections with the lines

$$L_a : \ \boldsymbol{x} = (1,0,0) + t(1,a,2), \qquad M_a : \ \boldsymbol{x} = (2,-1,0) + t(1,a,2), \qquad t \in \mathbb{R}.$$

Give the intersection parameters whenever they are unique. Include all cases in one table.

Exercise 2 (The perpendicular foot). Find the closest point to $Q = (3,-1,1)$ on the plane $2x + y - z = 3$ and the distance from Q to the plane. Repeat the computation with the equivalent equation $-4x - 2y + 2z = -6$. Check directly that the foot is on the plane and that the error is normal.

Exercise 3 (An intersection of planes and a parallel case). Find parametric equations for the common points of $2x + y - z = 3$ and $x - y + z = 0$. Check the line direction by crossing their normals. Then replace the second plane by $4x + 2y - 2z = k$, with $k \in \mathbb{R}$, and classify the common set for every k.

Exercise 4 (Coordinates and all exceptional angles). Give cylindrical and spherical coordinates of $(-\sqrt{3}, -1, -2)$ in Lecture 14's standard ranges. Use cylindrical r and spherical ρ. Also give every representation in those ranges of $(0,0,-3)$ and of the origin. Keep θ polar and φ azimuth.

Exercise 5 (Bounds for a region). Let the region $E \subset \mathbb{R}^3$ be given by

$$E = \{(x,y,z) : x^2 + y^2 + z^2 \le 9,\ z \ge \sqrt{x^2 + y^2},\ x \le 0\}.$$

Give cylindrical bounds and spherical bounds whose coordinate images are exactly E. Explain how your bounds cover the axis and origin. The task asks for a description of the points, not every redundant angle representation of those points. No integration is required.

Exercise 6 (A centred ellipse and its interior). The ellipse has Cartesian equation $x^2/9 + y^2/4 = 1$. Find its polar radius as a function of $0 \le \varphi < 2\pi$. Find the largest and smallest radial distances, including all angles at which they occur. Give polar bounds whose image is the closed elliptical region $x^2/9 + y^2/4 \le 1$.

Exercise 7 (The distance between two skew lines). Consider the spatial lines

$$L_1 : \boldsymbol{x} = (1,0,1) + s(1,1,0), \qquad L_2 : \boldsymbol{x} = (0,2,3) + t(1,0,1), \qquad s,t \in \mathbb{R}.$$

Show that they are *skew*: they are neither parallel nor intersecting. Find their distance, meaning the smallest distance between one point on each line, and find endpoints of a segment attaining that distance. Also find the distance from $Q = (0,2,3)$ to L_1, using the cross-product area formula, and find its perpendicular foot on L_1.